

Random Forest as a Novel Machine Learning Approach to Predict Landslide Susceptibility in Kalutara District, Sri Lanka

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Abstract—Prediction of landslide susceptibility is of vital importance as a research topic in the field of disaster management. Landslides are considered as natural disasters that frequently occur in Sri Lanka. This study presents an approach to predict the susceptibility of occurrence of landslides in Kalutara district, Sri Lanka, by integrating geo-spatial technologies such as Geographical Information Systems (GIS) together with machine learning to get a high performing prediction model with better accuracy. A landslide inventory of 84 landslides occurrences accompanied by a spatial database carrying data related to 12 landslide conditioning factors in Kalutara District, namely as aspect, slope, geology, land use, land-form, hydrology, Topographic Wetness Index (TWI), Stream Power Index (SPI), Sediment Transport Index (STI), soil type, soil thickness, and rainfall were employed in this study. A separate pilot study was executed to compare and analyze the correlation of landslide occurrences to landslide conditioning factors, and thus filter out the most appropriate set of conditioning factors for the given spatial context. The Random Forest (RF), a supervised machine learning-based classification algorithm, was employed to implement the landslide susceptibility prediction model.

The performance of the model was assessed and evaluated using accuracy, sensitivity, specificity, Receiver Operating Characteristic (ROC), and Kappa index. The results indicated 80.39% of accuracy, 84.00% of Sensitivity, and 76.92% of Specificity. While the Kappa index value was indicated as 0.61, the area under the ROC curve showed 79.46% of robustness for the model implemented.

Index Terms—Landslide Susceptibility Prediction, Landslide Conditioning Factors, GIS, Random Forest (RF)

I. INTRODUCTION

A definition for the term "Landslide" can be expressed as, a movement of a mass of rock, debris, or earth towards a slope [1]. Having considered as one of the most frequently occurring natural hazards [2] in Sri Lanka, about 30% of Sri Lanka's land area covering 10 districts [3] has been identified as prone to landslides.

Kalutara District positioned with-inside the Western Province of Sri Lanka between latitude 6°34'59.99" N and longitude 80°9'60.00" E. Kalutara district is among the districts having the highest frequency [3] of landslides. Therefore the efforts in the prediction of landslide susceptibility have been a major concern and important aspect in disaster management and ensuring the sustainability of developments in the district.

With soft computing approaches for landslide susceptibility prediction becoming popular in the world, different models have been implemented using machine learning techniques in different regions of the world showing better accuracy than statistical approaches [4]. The efficiency of the landslide susceptibility mapping and prediction methods depends on the method employed and the quality of the conditioning factors [5]. A wide range of conditioning factors could cause a landslide occurrence [6]. Despite, in a given study area, only some classes of landslide conditioning factors [7] will exert a substantial impact on the occurrence of a landslide. Therefore, deciding on the maximum suitable set of conditioning elements for a given examine location turns into greater importance for landslide studies [8].

For this study, contour data and geo-spatial statistical data were procured from the National Building Research Organization (NBRO), Sri Lanka, while precipitation data were procured from the Department of Meteorology, Sri Lanka. Based on the past research work [8], [9], [10], [11] and guidance received from the NBRO, initially 12 conditioning factors including aspect, slope, geology, land use, land-form, hydrology, Sediment Transport Index (STI), Stream Power Index (SPI), Topographic Wetness Index (TWI), soil type, soil thickness, and rainfall were selected. To select the most appropriate set of conditioning factors for the subjected area, a preliminary set of 12 conditioning factors had been analyzed in advance for the use of the Frequency Ratio and Information Gain Ratio. During the time frame of this study, Random Forest (RF) has not been widely explored in the Sri Lankan context as a novel machine learning approach for landslides susceptibility prediction. Therefore, this paper presents a research work to predict the susceptibility of landslide occurrence in Kalutara district, Sri Lanka by employing RF as the machine learning algorithm.

The rest of the sections of this paper are organized as follows. Section II unveils the past research work related to landslide studies. Section III describes the selected study area for this research. Section IV uncovers the methodological approach that was followed throughout this research study while Section V discusses the results obtained from the pilot study and evaluation of the model. Section VI provides the conclusion of this research work.

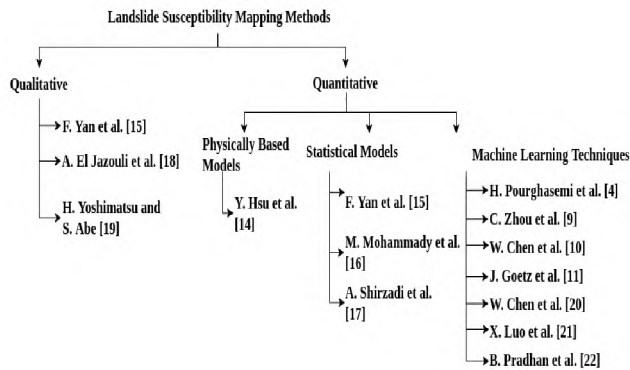


Fig. 1. Taxonomy of Methods of Landslide Susceptibility Mapping

II. RELATED WORK

In the research literature of landslide susceptibility mapping, two main approaches are there, as qualitative and quantitative [12]. Experts' opinions are the key foundation for qualitative methods [4]. Quantitative methods are mainly based on reason-cause relationships between landslides and related controlling factors [13]. A taxonomy of methods of landslide susceptibility mapping by H.R. Pourghasemi et al.[4] is illustrated in Fig. 1.

Physically-based models [14] require detailed site-specific geological data [4]. When analysing natural hazards, traditional statistical models [15], [16], [17] assume a suitable structural model at first, and then work on parametrizing that model. Therefore, the classification of each landslide conditioning factor becomes a crucial point contributing to the quality of the generated landslide susceptibility map [4]. In machine learning-based approaches, a set of powerful data-driven tools are employing learning algorithms to reveal the link between landslide related predictors and landslide occurrences. Thus, machine learning-based approaches overcome the restrictions related to data-dependent bi-variate, multi-variate statistical methods [4]. In qualitative studies, [15], [18], [19], a group of conditioning factors are examined and then weighted using methods such as Analytic Hierarchy Process (AHP) based on particular importance.

Table I demonstrates a comparison among the previous studies carried out in China, Malaysia, and Iran which have used different machine learning algorithms in landslide susceptibility prediction and the algorithm which reaped the highest accuracy from the employed algorithms. Through the literature, it is evident that RF has shown about 75%-84% [4], [10], [20] accuracy in the prediction of landslide occurrences in these regions. The majority of the studies have utilized about 12 landslide conditioning factors including slope, aspect, landform, geology, distance to rivers, STI, NDVI, distance to roads, SPI, profile curvature, plan curvature, TWI, etc [9], [20], [21] where methodologies such as information gain ratio, VIF, and tolerances [4] have been utilized to eliminate the less appropriate factors for the study. Friedman test, ROC, sensitivity, and specificity are among the performance

TABLE I
SUMMARY OF LITERATURE REVIEW

Paper	Algorithms Used	Number of Factors	Best Performer/ Reported Accuracy(AUC)
[4]	GAM, SVM, ANN, BRT, CART, GLM, QDA, MARS, RF, Naive Bayes	9	RF-83.7%
[9]	SVM, ANN, LR	12	SVM-0.912(Colluvial), SVM-0.932(Rockfall)
[10]	LMT, RBF, Bayes' Net	12	RF-0.752
[11]	SVM, RF, BPLDA, GLM, GAM, Weights Of Evidence(WOE)	12	GAM
[20]	RF, CART, LMT	12	RF-0.781(Prediction Rate), RF-0.837(Success Rate)
[21]	SVM, ANN, IVM	16	ANN-0.832
[22]	SVM, ANFIS, DT	8	ANFIS-94.21

measures used in past studies to analyze the results of the machine learning models.

There is a lack of studies carried out in landslide prediction using machine learning algorithms in the Sri Lankan context. The few types of research [3], [23], [24] carried out in Sri Lanka in this domain have explored landslide susceptibility mapping using LR, ANN, SVM, and NB employing 5 to 6 factors including geology, landform, land use, etc. The possibility of accompanying factors such as slope, aspect, SPI, etc. have not been considered. Comparative studies [16], [11] carried out in other regions of the world using machine learning techniques and statistical approaches for prediction of landslides have emphasized that machine learning approaches give better performance than statistical approaches. Benefits of using advanced machine learning approaches such as Random Forest, Artificial Neural Networks, Naive Bayes, Radial Basis Classifier, Decision Trees, etc. are not sufficiently reaped by the minimal set of research carried out in Sri Lanka on the subject of landslide prediction.

III. METHODOLOGY

The methodology in this study employed a hybrid approach that combined the design science research approach [25] and the quantitative research approach [26].

A. Landslide Inventory Map, Training and Testing Data Sets

The implementation of the landslide inventory map is the foremost step in landslide susceptibility prediction. It depicts the spatial dispersion of landslide events in a specific region over time. This map was created using the landslide database acquired from the NBRO. In this study, a total of 84 landslides that took place from 1984 to 2018 in Kalutara District, were analyzed. The center of the initiation area of each landslide was used to represent the landslide locations. Additionally, 84 non-landslide locations were generated from the landslide free area in Kalutara district by using a random sampling mechanism. For the susceptibility analysis, 70% of the entire data set was used for training and 30% was used for testing. This separation was also carried out randomly.

B. Landslide Conditioning Factor Maps

The landslide conditioning factors selected for the study are TWI, rainfall, geology, land use, slope, STI, aspect, soil thickness, hydrology, soil type, landform, and SPI.

The Slope is the rise or fall of a land surface. Aspect provides the directions that slopes face. The distance to waterways in the study area is given by hydrology. SPI measures the erosiveness of water flow in a specific catchment area. TWI is the quantifies water accumulation at a site. Slope failure and deposition are described by STI. Landforms are natural terrain features on the earth's surface. A soil type is a taxonomic unit in soil science. The thickness of the soil is the distance between the soil surface and the underlying bedrock. Land use involves modification of the natural environment into the built environment. Geology is concerned about the materials of which earth is made. Since most of the landslides in Kalutara was caused due to heavy rainfall, it was treated as the triggering factor for the landslide occurrence in this study.

A contour map of Kalutara was utilized to generate the Digital Elevation Model (DEM) with a grid size of 10X10 m. DEM data was used to create raster maps of TWI, STI, SPI, aspect, and slope. Thematic maps for geology, land use, landform, hydrology, soil type, and thickness were generated in vector format using the geospatial statistical data. The average annual rainfall data collected from 26 rain gauge stations in Kalutara district was utilized to generate the rainfall map. The raster maps generated for rainfall, slope, aspect, STI, TWI, and SPI were classified into five equal interval classes each.

C. Correlation Analysis

The Frequency Ratio is employed in evaluating the relationship between landslide events and their conditioning factors. It is considered as the proportion between the total area of a landslide that occurred of a particular class for a given condition factor, to the entire subjected area, as a percentage [16].

D. Landslide Conditioning Factor Selection

Noise present in features can curtail the reliability of the model performance and its accuracy. Therefore, selecting conditioning factors with high predictive ability is important [8]. Information Gain and Information Gain Ratio has been widely used in landslide studies for this purpose. Information Gain Ratio eliminates the natural bias in Information Gain. Therefore, quantification of predictiveness of causative factors was carried out using Information Gain Ratio so that the factors with null predictiveness can be removed. Thus increasing the reliability of the prediction model.

The Information Gain Ratio is obtained by the ratio between information gain to the innate information. Factors with higher information gain ratio indicate a higher contribution to the prediction [8].

E. RF-based Prediction Model for Landslide Susceptibility

The proposed approach with Machine learning in this study, provide new analytical and modeling capabilities when combined with the processing capabilities of GIS and data set with landslide inventory and conditioning factors [11]. In this study, a susceptibility prediction model targeting the Kalutara district, Sri Lanka, was implemented utilizing the RF algorithm.

RF is a novel ensemble classifier that employs a forest of decision trees to make the final prediction [27]. In the training phase, each tree in the forest is grown by randomly selecting samples(with replacement) from the training data set. This results in training each tree using different sets of samples from the training data set. Therefore, the ensemble prediction resulted from the forest of trees tends to be more accurate than an individual prediction from a single decision tree [27]. In the testing phase, the final prediction is made by averaging the prediction of each decision tree.

In RF nodes in each tree is split considering a random subset of all features available. From the feature subset, the feature that produces the most separation between the data in the left node and the right node is selected. This random feature selection helps to reduce the out-of-bag error of a given tree while reducing the correlation between two generated decision trees [27].

RF Classifier available in Scikit Learn [28] was used to fit the data to the model. A data set containing 168 landslide and non-landslide events with their corresponding values for the 12 landslide conditioning factors was used to train(70%) and test(30%) the RF model. The values of conditioning factors at each landslide and non-landslide point were extracted by applying raster and vector functions in QGIS on the thematic maps.

F. Performance Evaluation of the Model

The performance measures for the RF-based prediction model were quantified using sensitivity, specificity, accuracy, kappa index, and the area under the ROC curve.

IV. RESULTS OF THE STUDY

This section unravels the obtained results from the pilot study and assessment on the performance of the RF-based prediction model.

A. Evaluation of pilot study

Results obtained from the pilot study assisted in improving the quality of the constructed feature pool and enhancing the reliability of the prediction model by identifying the highly correlated classes of conditioning factors with landslide occurrence, identifying conditioning factors with null predictiveness, and elimination of them. Table II demonstrates the conditioning factor classes along with the frequency ratio values obtained. $FR > 1$ suggests a strong correlation indicating high susceptibility to landslides and $FR < 1$ indicate lower susceptibility. An average value is represented when $FR = 1$.

TABLE II
FREQUENCY RATIO OF CONDITIONING FACTORS

Attribute	Classes : FR			
Geology	QtzFd	2.29	Qtz	1.55
	ChGn	1.07	GrGn	0.07
	GtBtGn	0	Ch	0
	Kh	0		
Soil Type	Colluvium	2.99	Alluvial	2.76
	Residual	1.02	Boulders	0.68
	Rock Exposure	0.35		
Soil Thickness	Rs t=4-5	0.597	Rs t=3-4	0.140
	Rs t=2-3	0.132	coll t=1	0.108
	coll t=2	0.024		
Landform	D38	0.107	D32	0.082
	D37	0.069	C22	0.043
	C23	0.036	C16	0.028
	D34	0.022		
Land Use	MTC ¹	0.005	Paddy	0.002
	AC ²	1.524	SL ³	0.053
	Rubber	0.006	Tea	0.052
	DF ⁴	0.051	DMF ⁵	0.058
	VH ⁶	10.583	Rubber/SL ³	1.253
Hydrology	33-270	4.235	270-573	1.447
	573-843	0.548	843-1113	0.115
	1113-1383	0		
Rainfall	2490-2940	0	2940-3390	0
	3390-3840	0.489	3840-4290	3.094
Slope	0-16	0.242	16-32	1.908
	32-48	2.179	48-64	1.739
	64-80	1.265		
Aspect	Flat	0.971	NorthEast	0.948
	East	1.012	SouthEast	0.974
	South	1.123	SouthWest	1.058
	West	1.046	NorthWest	0.929
	North	0.828		
SPI	(-2480)-(-1719)	0.378	(-1719)-(-958)	2.144
	(-958)-(-197)	1.933	(-197)-564	0.970
	564-1325	0.918		
TWI	8-10	2.039	10-12	1.682
	12-14	0.502	14-16	0
	16-18	0		
STI	0-5	0.887	5-10	1.287
	10-15	1.690	15-20	1.739
	20-25	2.579		

¹ MTC:- Land Use: Mixed Tree Crops

² AC:- Land Use: Annual Crops ³ SL:- Land Use: Scrub Land

⁴ DF:- Land Use: Degraded Forest ⁵ DMF:- Land Use: Dense Mixed Forests ⁶ VH:- Land Use: Village Home-lets

Slope angles 0°-16° showed a frequency ratio of 0.242, which suggests a low chance of landslide occurrence, while the slope angle 32°-48° obtained the maximum ratio value indicating the highest correlation. The probabilities of landslide occurrence in aspects facing south showed the highest frequency ratio of 1.123. North facing aspects (FR=0.828) demonstrates the least susceptibility to landslides. The SPI values (FR=0.378) ranging from (-2480)-(-1719) are least susceptible to landslides and the SPI values (-1719)-(-958) with an FR value of 2.144 given the highest correlation with landslide occurrence. The TWI between 8 and 12 has the highest susceptibility to landslides with FR values 2.039 and 1.682. TWI values greater than 14 showed the least probability of landslide occurrence.

Colluvium soil demonstrated the highest likelihood of a landslide occurrence than that of the other soil types bearing the highest FR (2.99). D38 and D32 landform types

TABLE III
INFORMATION GAIN RATIO OF CONDITIONING FACTORS

Factor	Information Gain Ratio
Slope	0.0615
Soil Type	0.0035
Soil Thickness	0.1568
Geology	0.1456
Landform	0.1010
Hydrology	0.0721
Land use	0.0250
STI	0.0184
SPI	0.1299
TWI	0.0582
Aspect	0.0362
Rainfall	0.0654

TABLE IV
CONFUSION MATRIX

True Labels	Predicted Labels	
	landslide	non-landslide
landslide	21	4
non-landslide	6	20

with FR values 0.107 and 0.082 are most probable to be subjected to landslides. Village home-lets with FR of 10.583 contribute the most towards landslides among land-use types. With FR value of 0.597, the residual soil type of 4-5m thickness contributes most for the landslide susceptibility, while Colluvium soil type-2m is at its lowest with FR of 0.024. The Quartzo Feldspathic (FR = 2.29) geology is most susceptible to landslides while hydrology indicates the highest susceptibility when the distance to waterways is between 33m to 270mm (FR=4.235). The rainfall between 3840mm to 4290mm demonstrates the highest probability of landslides with a frequency ratio of 3.094.

Table III lists the computed information gain ratios for the 12 factors employed. Soil thickness (0.1568) demonstrated the highest information gain ratio. Geology (0.1456), SPI (0.1299), landform (0.1010), hydrology (0.0721), rainfall (0.0654) and slope (0.0615) showed higher gain ratios. The minimum information gain ratios were demonstrated by TWI (0.0582), land use (0.0250), aspect (0.0362), STI (0.0184), and soil type (0.0035). It was concluded that zero or null predictiveness to landslide events was not demonstrated by any of the conditioning factors since none of the factors showed zero gain ratio value.

From the results obtained for the pilot study, all 12 factors were chosen to implement the RF prediction model based on their correlation and predictive capability to landslide occurrence.

B. Evaluation of Landslide Susceptibility Prediction Model

The classifier with optimal hyper-parameters was employed to fit the training data and it was used to predict the landslide susceptibility using the test data. Table IV demonstrates the confusion matrix generated from predictions obtained through the application of RF classification on the test data set.

The Confusion matrix was used to calculate the sensitivity, specificity, and accuracy values for the final model. Table V outlines the above-mentioned performance measures.

TABLE V
MODEL PERFORMANCE MEASURES

Performance Measure	Value Obtained
Accuracy	80.39%
Sensitivity	84.00%
Specificity	76.92%
Kappa Index	0.61

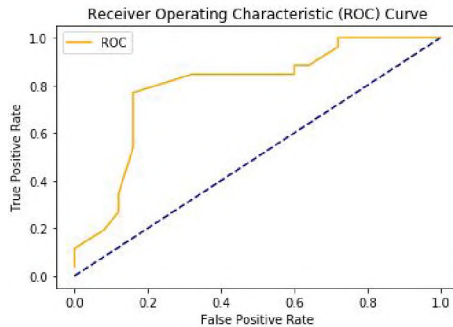


Fig. 2. Receiver Operating Characteristic

The model showed 80.39% accuracy on testing data. Out of the 51 test instances, 21 true positive classifications, and 20 true negative classifications were given by the model. This indicates that the model was able to correctly identify the true landslide and non-landslide points the study area with confidence over 80%. Meanwhile, a total of 10 instances were misclassified including, 4 false positives and 6 false negatives out of 51. The Kappa index value of 0.61, indicated that the proposed approach shows nearly substantial reliability. However, landslide points classified as false negatives and false positives are relatively low. Kappa index shows nearly the substantial reliability of the model.

The ROC curve (Fig. 2) was used to evaluate the prediction capability of the model. The area under the ROC curve obtained a value of 0.7946 (79.46%) which indicated a substantial agreement between the true landslide points and the predicted landslide points. Since the AUC value was between 0.7 to 0.8 for the final model, it informs a "good" predictive capability of the model.

The performance of the model was optimized by tuning the number of decision trees. The optimal number of decision trees was experimentally identified by testing model performance for different numbers of trees. Fig. 3 demonstrates the variation

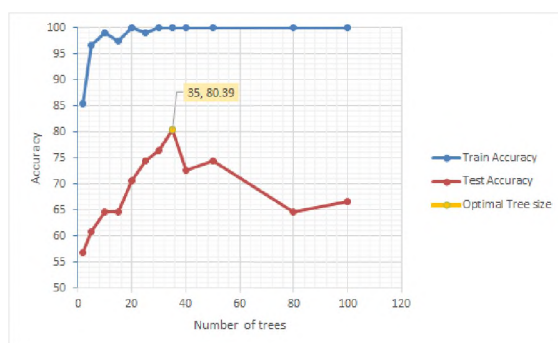


Fig. 3. Number of Trees Vs Model Accuracy Graph

in the training accuracy and the test accuracy when the number of decision trees of the classifier is changed. The best model performance was achieved when the number of decision trees was equal to 35. The model indicated 100% of training accuracy and 80.39% of testing accuracy. The performance of the model continued to show an improvement when the number of decision trees was altered from 1 to 35. When the number of decision trees was incremented more than 35, the testing accuracy decreased while the training accuracy remained 100%.

The results obtained from the RF-based landslide prediction model were used to implement a "Landslide Susceptibility Map" (LSM) for the study area, Kalutara district. The Landslide Susceptibility Indices (LSIs) were generated and reclassified according to different susceptibility classes. The trained model was used in the calculation of the LSIs and the indices were reclassified into 5 susceptibility classes as very high, high, moderate, low, and very low, using the natural breaks method [29]. The landslide susceptibility map generated for the Kalutara district is shown in Fig. 4.

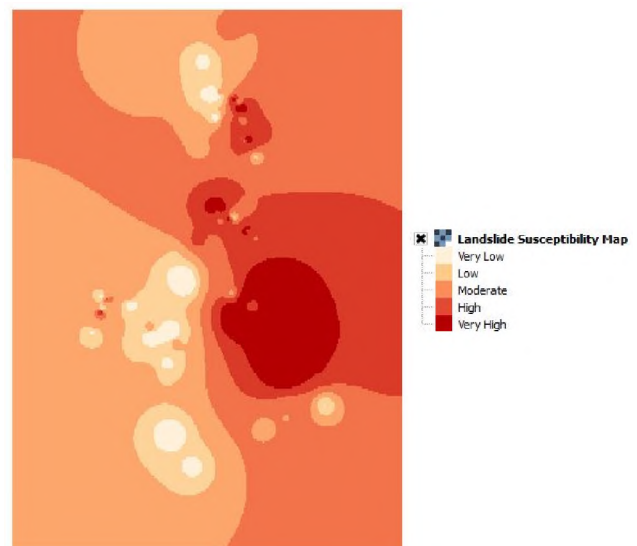


Fig. 4. Landslide Susceptibility Map

V. CONCLUSION

This paper contributes to predicting landslides susceptibility in Kalutara District, Sri Lanka. The study employs 12 selected terrain and triggering factors of landslide occurrence, and uses RF as the machine learning algorithm for the susceptibility mapping.

Results obtained from the frequency ratio analysis indicated that not all the classes of conditioning factors affect directly the occurrence of landslides while certain classes of them giving a major impact. None of the parameters out of 11 landslide terrain factors considered in the study, demonstrated zero or null predictiveness with relating to the occurrence of a landslide, according to the information gain ratio values calculated. Therefore all 12 conditioning factors including both terrain and triggering factors were employed to implement the model.

The final landslide susceptibility prediction model implemented using RF demonstrated an accuracy of 80.39%, while indicating 84.00% and 76.92% for sensitivity, and specificity respectively on test data. AUC value was 0.7946 (79.46%) indicating a "good" predictive ability for the model. Kappa index value of 0.61, suggested nearly substantial reliability of the model for the data set.

As for future improvements, the implemented model can be expanded to investigate other landslide-prone locations in Sri Lanka and its performance can be assessed by accuracy on identifying previous landslide locations in those areas. Also, it can be trained to locate new landslide-prone areas in those districts. The methodology followed in this study should be fine-tuned with bias-corrected RF methods and improved bagging techniques to address the bias and variance problems associated with RF. Since current the model is dealing with a context-specific data set, it's important to maintain the bias and the variance at a level that they are not improved at an excessive expense to the other.

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